

NCVET 2025

Assessment Agency Technology & Digital Infrastructure Requirements

Technology, Digital Infrastructure, Assessment Engine, Portal and Data Management

SOURCE

NCVET — Guidelines for Recognition and Regulation of Assessment Agencies, June 2025.

Scope control: This document contains only the technology and digital-infrastructure requirements stated in the supplied NCVET Assessment Agency Guidelines 2025. No additional technology features have been added beyond the supplied source.

Prepared as a professional reference document

TECHNOLOGY, DIGITAL INFRASTRUCTURE, ASSESSMENT ENGINE, PORTAL AND DATA MANAGEMENT REQUIREMENTS

Source

NCVET — Guidelines for Recognition and Regulation of Assessment Agencies, June 2025.

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1. DIGITAL INFRASTRUCTURE REQUIRED FOR ASSESSMENTS

As per Section 2.3.6.2 of the NCVET AA Guidelines 2025:

Entities seeking recognition as Assessment Agencies must have a strong digital infrastructure to support secure and efficient assessments.

The Guidelines specifically mention:

- AI-enabled test engines
- Online proctoring systems
- Performance analytics

These are intended to support transparency and scalability of the assessment process.

2. ASSESSMENT TOOLS AND ASSESSMENT MODES

The Assessment Agency must have adequate assessment tools and aids to support delivery of assessments pertaining to the qualification.

The Guidelines specify support for:

- Offline assessment
- Blended assessment
- Online assessment
- Proctored assessment

The assessment infrastructure must therefore support the assessment format applicable to the relevant qualification.

3. AI-ENABLED ASSESSMENT ENGINE

The AA must have state-of-the-art technology and an AI-enabled test engine capable of delivering and monitoring online and blended assessment lifecycles.

The system should provide:

- Assessment lifecycle dashboards
- Performance analytics for stakeholders
- Systems to refresh assessment content periodically

The assessment engine must support the assessment lifecycle and monitoring of online and blended assessments.

4. AUTOMATED CREATION OF TESTS

The Assessment Engine and Portals used for the assessment process must provide automated creation of tests from the question bank.

Tests created for each qualification should follow assessments mapped to:

Qualification → NOS → Performance Criteria (PC)

The system must follow a standardized pattern for question selection while ensuring:

- Balanced difficulty levels
- Weightage distribution according to Bloom's Taxonomy

5. RANDOMIZED QUESTION SELECTION

The assessment engine should incorporate an algorithm that selects questions in a randomized order.

The randomized selection must ensure:

- Balanced distribution of difficulty levels
- Appropriate weightage according to Bloom's Taxonomy

The difficulty-level weightage for each assessment is to be determined by the AA in consultation with the concerned AB, ensuring alignment with the assessment methodology and intended Learning Outcomes (LOs).

6. ONLINE PROCTORING / AUTO-PROCTORING

The Guidelines require audio and video systems for:

- Online proctoring
- Auto proctoring

These systems are to record the ongoing assessment at assessment centres.

The Assessment Engine/Portal should therefore support recording of the ongoing assessment through the specified audio/video systems.

7. LIVE PROCTORING AND ANTI-MALPRACTICE

The online assessment platform should incorporate real-time proctoring that can be monitored throughout the assessment process, from the beginning until the end.

The platform should also be capable of:

- Recording ongoing assessment activity
- Detecting/identifying suspicious behaviour
- Recording malpractice activity
- Flagging malpractice activity

The Guidelines specifically state that, in case of malpractice by a learner, the portal should be able to record and flag that activity.

8. REAL-TIME PHOTO CAPTURE

When an assessment begins, the smart AI-enabled platform should provide:

- Live video feeds
- Random photo capturing at regular intervals

These facilities are intended to assist in detecting suspicious behaviour of the candidate.

9. GEO-TAGGING

The technology infrastructure must provide geo-tagging facilities for:

- Assessors
- Proctors

The purpose is to verify their presence at the training/assessment centres.

The Assessment Tracking requirement further specifies that when an Assessor logs in or logs out of the Assessor App:

- Geo-location should be recorded

- Date should be recorded
- Time should be recorded

This recording should also work in offline mode.

The same tracking is required for students.

The Guidelines also state that details are to be sent by email to the Awarding Bodies at the time of Assessor Login.

10. PERFORMANCE REPORTING AND ANALYTICS

The technology system must provide performance reporting and analytics.

The portal should be capable of providing performance reports and analytics.

The Guidelines specifically require graphical performance analysis based on five parameters:

1. Assessor
2. Training Partner
3. Job Role
4. State
5. Date — From and To

11. ONLINE VERIFICATION AND AUTHENTICITY OF CANDIDATES

The digital infrastructure must provide online verification and authenticity of candidates undergoing assessments.

This is specifically mentioned with reference to:

- Auto online proctoring
- Geo-tagging

The purpose is to support verification and authenticity of candidates undergoing assessment.

12. ONLINE AND OFFLINE MODE

The Skill Assessment Platform must work in:

- Online mode
- Offline mode

Offline functionality is specifically required where there are connectivity issues.

The Guidelines therefore explicitly require the assessment platform to support assessment operations even where connectivity is unavailable or inadequate.

13. QUESTION DIFFICULTY CLASSIFICATION

The platform should automatically classify questions according to difficulty levels.

The Guidelines specify classification into:

- Easy
- Medium
- Tough

The platform should also provide a ready-reference report showing the usage of each question.

For questions that are not answered correctly by most students, feedback may be provided to:

- Awarding Body (AB)
- Training Partner

14. QUESTION BANK REQUIREMENTS

The AA must have question banks.

The Guidelines specify that the organization should have question banks for at least 5% of the qualifications available in the relevant sector as listed on the National Qualification Register (NQR), for which recognition is being sought on the date of application.

The AA must prepare multiple questions for each Performance Criteria (PC).

The minimum average number of questions per PC within a qualification / standard NOS / micro-credential / nano-credential is:

At least 5 questions per PC.

Question banks must be prepared in consultation with the concerned Awarding Body to ensure the intended outcomes.

15. QUESTION BANK DIFFICULTY AND BLOOM'S TAXONOMY

Question banks should have a balanced mix of difficulty levels:

- Low
- Medium
- High

They should cover different types of questions following Bloom's Taxonomy.

The AA is responsible for periodic review of question banks.

Each review cycle should be conducted in consultation with the concerned AB.

Question banks must be changed/upgraded periodically.

Question banks and assessment tools should also be available in other Indian languages relevant to the region seeking recognition.

16. TYPES OF QUESTIONS SUPPORTED

The AA's assessment engine should be able to handle:

- Multiple-choice questions
- Short-answer questions

The Guidelines also encourage AAs to include scenario-based questions in question banks.

Depending on the qualification, assessment may also be conducted online/offline using simulators, as specified in the applicable blended learning guidelines.

The AA may also create question banks for:

- Viva questions
- Practical assessments

17. ASSESSMENT DATA AND DIGITAL STORAGE

The portal should be capable of storing:

- Videos pertaining to assessments
- Other assessment-related data

This storage is required as part of the governance mechanism.

18. DATA MANAGEMENT SYSTEM (DMS)

The Assessment Agency must have a Database Management System (DMS) and Scientific Database Management System (SDMS) for the assessment process.

These systems must have:

A. Recording, Validation and Reporting

Systems to:

- Record relevant assessment details
- Validate relevant assessment details
- Report relevant assessment details

The purpose is to ensure governance of the assessment process.

B. Information Security

There must be clearly documented procedures for security of information relating to learner details, including learner results.

C. Data Collection

Tools and relevant software for data collection must be available.

D. Data Analysis

Management and learning-support staff must have the necessary skills to analyse collated assessment data and provide feedback for improving operations.

E. Data Protection

The entity must adhere to data maintenance and protection requirements under the:

Digital Personal Data Protection Act, 2023.

19. DATA SECURITY

The Skill Assessment Platform must ensure security of data.

The Guidelines state that the platform should ensure zero possibility of content theft and perform suggested information-security audits at regular intervals in accordance with Government of India guidelines.

20. STUDENT DATA PRIVACY

Privacy of student data must be maintained according to the guidelines of the Government of India.

The AA must also maintain the security of learner information, including results.

Data maintenance and protection must comply with the Digital Personal Data Protection Act, 2023.

21. USER-FRIENDLY INTERFACE

The portal should have a user-friendly interface.

It should:

- Be easy to access
- Be easy to operate
- Not require prior expertise
- Be designed for ease and accuracy of use

22. WEBSITE REQUIREMENT

The AA must have a fully functional website.

The website must provide the following critical information:

A. Operational Team

- Details of operational teams
- Organogram

B. Assessor and Proctor Information

- Assessor details
- Proctor details
- Educational qualifications
- Relevant experience

C. Sample Assessment Papers

Sample assessment papers for various qualifications relating to the domain and sector for which recognition is sought.

D. Industry Linkages

Information on industry linkages to strengthen assessment delivery.

E. Grievance and POSH

Information on:

- Grievance redressal
- POSH mechanisms

F. Registered Office Documents

Availability of supporting documents of the registered office, such as:

- Ownership document
- Lease agreement

G. Commercial Premises Proof

Proof that the premises are being used for commercial purposes, such as:

- Telephone bill
- Internet bill
- Electricity bill

23. ICT TOOLS FOR PERSONS WITH DISABILITIES

For assessment delivery to learners with disabilities, the Guidelines specify the following essential markers:

A. Special Instructors / Content Developers

Availability of special instructors/content developers for creating specific disability-based assessment content.

B. Trained Proctors and Assessors

Availability of trained proctors and assessors who understand and can align with PwD learners.

C. ICT Tools

Availability of ICT tools for:

- Delivering assessments
- Monitoring assessments

D. Result Publishing Tools

Availability of tools to publish results in line with the needs of Persons with Disabilities.

24. SIMULATORS

Where blended mode is chosen, the AA must have availability of:

- Simulators
- Other tools for assessment

where applicable to the assessment.

25. ASSESSMENT DATA REPOSITORY

The AA must maintain a repository of learners' assessment data.

This may be maintained:

- Through its own in-house team; or
- Through an agency proposed by NCVET

The repository must follow Government of India requirements relating to:

- Data security
- Data privacy
- Data storage

The AA must also maintain a repository of assessors containing:

- Their qualifications
- Number of years of work experience

26. ASSESSOR APP — REQUIREMENTS ACTUALLY STATED IN THE GUIDELINES

The Guidelines refer specifically to an Assessor App in relation to assessment tracking.

At Assessor Login/Logout, the system must record:

- Geo-location
- Date
- Time

This must be capable of being recorded even in offline mode.

The system must also track the students.

The Guidelines further state that details are sent by email to the Awarding Body at the time of Assessor Login.

Important: The Guidelines do not provide a complete separate specification for a mobile application. Therefore, features beyond these stated requirements should not be treated as NCVET-mandated features solely on the basis of these Guidelines.

27. TECHNOLOGY-RELATED MONITORING REQUIREMENTS

The NCVET Excellence-Risk Framework includes technology-related assessment-delivery parameters.

These include:

- Availability of data-storage mechanism as per the Digital Personal Data Protection Act, 2023
- Video evidence of assessment captured on the AA assessment engine/portal/app
- Offline assessment conducted by a certified Assessor
- Result submission within the prescribed Turnaround Time
- Process to ensure that one assessor cannot assess two or more batches on the same day at different locations

These parameters are part of the monitoring and evaluation framework.

28. CONSOLIDATED TECHNOLOGY REQUIREMENTS

Based strictly on the NCVET AA Guidelines 2025, the technology ecosystem should cover the following areas:

ASSESSMENT PLATFORM

- AI-enabled test engine

- Automated test creation
- Question-bank integration
- NOS/PC mapping
- Randomized question selection
- Difficulty-level balancing
- Bloom's Taxonomy weightage
- Online assessment
- Offline assessment
- Blended assessment
- Proctored assessment

PROCTORING

- Online proctoring
- Auto proctoring
- Real-time monitoring
- Audio recording
- Video recording
- Live video feed
- Random photo capture
- Anti-malpractice detection
- Malpractice recording and flagging

ASSESSOR / LOCATION TRACKING

- Assessor App login/logout tracking
- Geo-location
- Date stamping
- Time stamping
- Offline geo-location recording
- Student tracking
- Email communication to AB at Assessor Login

ANALYTICS

- Performance reports
- Performance analytics
- Graphical performance analysis
- Assessor-wise analysis
- Training Partner-wise analysis
- Job Role-wise analysis
- State-wise analysis
- Date-wise analysis

QUESTION BANK

- Minimum 5 questions per PC on average
- Question banks for at least 5% of applicable sector qualifications at application stage
- Multiple-choice questions
- Short-answer questions
- Easy/medium/hard questions
- Bloom's Taxonomy
- Periodic review
- Periodic upgrading

- Indian-language question banks as required
- Viva questions
- Practical assessment questions
- Scenario-based questions encouraged

DATA MANAGEMENT

- DMS
- SDMS
- Assessment-data recording
- Data validation
- Data reporting
- Data collection tools/software
- Data analysis
- Learner-result security
- Learner-data repository
- Assessor repository
- Video/data storage
- Data privacy
- Data protection
- DPDP Act, 2023 compliance

WEBSITE

- Operational team details
- Organogram
- Assessor details
- Proctor details
- Qualifications and experience
- Sample assessment papers
- Industry-linkage information
- Grievance-redressal information
- POSH information
- Registered-office supporting documents

PwD / ACCESSIBILITY

- Disability-specific assessment content
- Special instructors/content developers
- Trained assessors/proctors
- ICT tools for assessment and monitoring
- Tools for publishing results according to PwD needs

OTHER TECHNOLOGY

- Simulators where applicable
- Assessment tools and aids
- Information-security audits
- Secure assessment content
- User-friendly interface

29. SOURCE LIMITATION

The above list is limited to what is stated in the NCVET Assessment Agency Guidelines 2025.

The Guidelines do NOT, in the cited technology section, prescribe a complete software-development specification such as:

- Specific programming language
- Specific mobile operating system
- Specific database technology
- Specific cloud provider
- Specific server architecture
- Specific API architecture
- Specific biometric technology
- Specific facial-recognition technology
- Specific GPS hardware
- Specific payment gateway
- Specific CRM
- Specific help-desk software

Therefore, these should not be represented as NCVET-mandated requirements unless supported by another applicable NCVET document, notification, guideline or AB requirement.

30. PRIMARY NCVET SECTIONS USED

The main source sections for this technology requirement document are:

Section 2.3.6.2 — Digital Infrastructure required for Assessments

Information Technology: Assessment Engine, Portals

Database Management System (DMS) and Scientific Database Management System (SDMS)

Section 4.1 — Development of Question Banks

Section 4.2 — Assessment

Section 5 — Monitoring and Evaluation / Excellence-Risk Framework

These sections should be referred to when preparing the AA's technology infrastructure and compliance documentation.